

S3 Sample: Structure Preservation

Section One - Retrieval Pipeline Overview

This document exercises the PDF normalization path with a real text layer. Headings at three sizes, a fenced-style monospace code block and a ruled table must all survive conversion into normalized markdown. Anchor: dense and sparse channels are fused by reciprocal rank fusion before threshold filtering.

Fusion and Threshold Details

```
def rrf_fusion(dense_hits, sparse_hits, k=60):
    scores = {}
    for channel in (dense_hits, sparse_hits):
        for rank, hit in enumerate(channel):
            scores[hit.id] = scores.get(hit.id, 0.0) + 1.0 / (k + rank + 1)
    return sorted(scores.items(), key=lambda kv: -kv[1])
```

Section Two - Channel Configuration

The two retrieval channels and their models are summarized below. Every cell text is a registered anchor substring for the sample manifest.

channel	model	purpose	status
dense	Qwen3-Embedding-0.6B	semantic recall	active
sparse	Qdrant/bm25 lexical	lexical recall	active
fusion	RRF k=60 fixed	rank merging	frozen

Closing body paragraph of S3. It exists to verify that ordinary prose after a table is not swallowed by table detection during normalization.